

Technical Design Report

Defence Drone Navigation using GBFS and A* Search

Course: ACI | Assignment: PS4 | Group: G116

1. Project Summary

An autonomous UAV navigates an 8×8 grid from Start (0,0) to Goal (6,7), avoiding No-Fly Zones (N) and minimizing Weather Hazard (W) penalties. Implemented: GBFS with h1 (Euclidean), GBFS with h2 (Bounding-Box Risk-Weighted), and A* with h1 for comparison.

2. Environment & Problem Modelling

Symbol	Description	Cost
S	Start Node (0,0)	1
E	Goal Node (6,7)	1
.	Safe Airspace	1
W	Weather Hazard	4
N	No-Fly Zone	Blocked

Environment Statistics: • Grid Size = 8 × 8 • No-Fly Zones = 8 • Weather Hazards = 7 • Start = (0,0) • Goal = (6,7)

3. PEAS Components

Component	Description
Performance	Minimize path cost; reach goal; avoid N zones; reduce W penalties
Environment	8×8 grid, static, fully observable, deterministic, single agent
Actuators	Orthogonal movement (N/S/E/W); navigation controller
Sensors	GPS position; grid map; obstacle/weather/goal detection

4. Alternate Modelling Approach

Primary model: 2D matrix grid with O(1) cell lookup. Alternate: explicit graph G=(V,E) where V = passable cells (56 nodes) and E = orthogonal edges with transition costs. Performance: matrix is faster for neighbour expansion on dense grids; graph model uses less memory on sparse/large maps and scales better when the grid is not rectangular. For this 8×8 assignment both are equivalent; matrix was chosen for simplicity.

5. System Architecture

Component	Choice	Reason
Environment	2D Matrix	O(1) access
OPEN	Priority Queue (heap)	Efficient min-h extraction
CLOSED	Hash Set	O(1) revisit check
Path	Parent dict	O(path) reconstruction

6. Heuristic Engineering & Execution Flow

Heuristic h1 – Euclidean Distance

$h1(n) = \sqrt{(x_r - x_g)^2 + (y_r - y_g)^2}$. Fast, goal-directed, admissible for A*.

Heuristic h2 – Bounding-Box Risk-Weighted

$h2(n) = \text{Manhattan}(n, \text{goal}) \times \text{avg}(\text{cell costs in bounding box from } n \text{ to goal})$. Future-aware: penalises paths through high-cost hazard regions. Example: $h2(3,3)=15.40$.

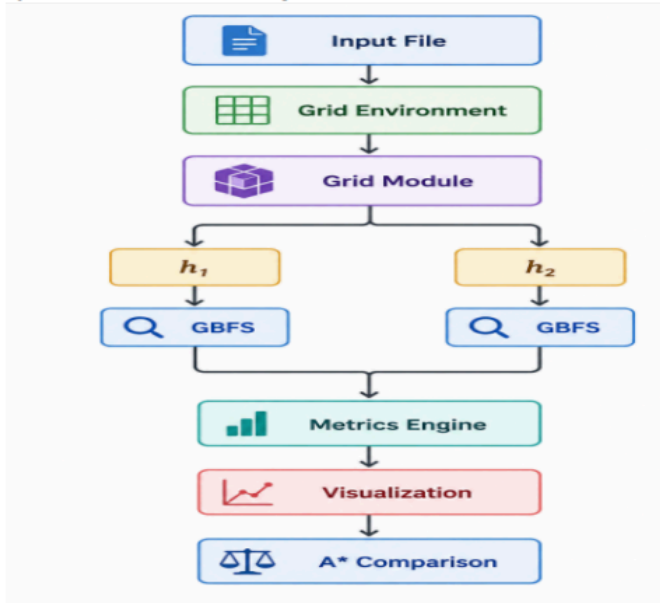
GBFS Algorithm

1. Pop min-h node from OPEN
2. If goal, stop
3. Add to CLOSED
4. Expand neighbours
5. Repeat

Execution Flow (Sequence Diagram)

1. Read inputPS04.txt → Load 8×8 grid, start (0,0), goal (6,7)
2. Run GBFS-h1 → Euclidean heuristic search, record path & metrics
3. Run GBFS-h2 → Bounding-box heuristic search, record path & metrics
4. Run A*-h1 → Optimal path benchmark with $f(n)=g(n)+h(n)$
5. Write outputPS04.txt → Metrics table, PEAS, trap check, complexity
6. Generate PNG charts → Grid paths, comparison, heuristic plots

System Architecture & Design Flow:



Data Structures & Tie-Breaking

Structure	Role	Complexity
heapq (OPEN)	Frontier priority queue	push/pop $O(\log n)$
set (CLOSED)	Explored nodes	add/lookup $O(1)$
dict (parent)	Path reconstruction	$O(1)$ lookup

Node priority on equal h: North > East > South > West (counter tie-break in heap).

7. Results & Comparison (outputPS04.txt)

Metrics Table:

Metric	Description	GBFS-h1	GBFS-h2	A*-h1
Nodes Expanded	Total explored nodes	14	14	29
Runtime	Execution time (ms)	0.139	0.191	0.2279
Memory Usage	OPEN + CLOSED size	23	28	40
Total Path Cost	Sum of transition costs	17	17	14
Path Length	Total moves	13	13	13
Heuristic	Heuristic value at start	9.2195	29.4821	9.2195

GBFS vs A* Comparison (Heuristic h1):

Criterion	GBFS (h1)	A* (h1)
Complete?	Yes (with closed set)	Yes
Optimal?	No	Yes
Nodes Expanded	14	29
Runtime (ms)	0.139	0.2279
Memory Usage	23	40
Total Path Cost	17	14
Path Length	13	13
Faster to goal?	Yes (fewer nodes, lower ms)	No (more exploration)

Trap Identification (Expectation 6)

Heuristic	H-Value	Trapped at Node	Iteration	Escaped To	Escape Iter.
—	—	No traps detected	—	—	—

Both GBFS heuristics found 13-move paths costing 17 (1 weather zone). A* found optimal cost 14 with zero weather penalties.

8. Analysis, Visualizations & Conclusion

Heuristic & Algorithm Comparison Charts



Complexity Analysis

Algorithm	Time	Space	Nodes	Notes
GBFS h1	$O(V \log V)$	$O(V)$	14	Fastest; not optimal
GBFS h2	$O(V \log V)$	$O(V)$	14	Risk-aware; higher h2 compute
A* h1	$O(V \log V)$	$O(V)$	29	Optimal path; more exploration

Heuristic Analysis

h1 (Euclidean): fastest runtime (0.139 ms), simple geometry, ignores hazard layout. h2 (Bounding-Box): higher start $h=29.48$, accounts for future hazard costs; same path cost on this grid. h1 is better for speed; h2 is better for hazard-aware routing on complex maps.

9. Conclusion

GBFS is complete with a closed set on finite grids but not optimal — it found cost-17 paths vs A* optimal cost-14. GBFS-h1 is fastest (14 nodes, 0.139 ms); A* guarantees optimality at higher cost (29 nodes, 0.228 ms). GBFS-h2 balances efficiency with risk awareness. Recommendation: A* for mission-critical optimality; GBFS-h1 for real-time response; GBFS-h2 for hazard-dense environments.